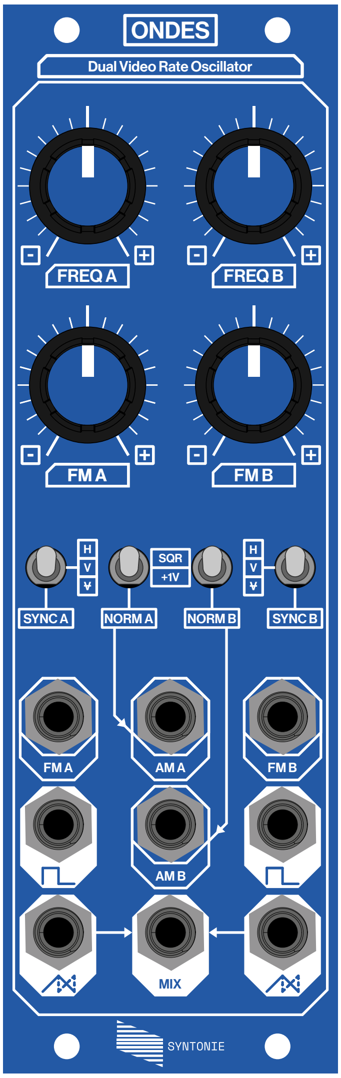
Ondes

Dual Video Rate Oscillator ¬
User documentation

SYNTONIE



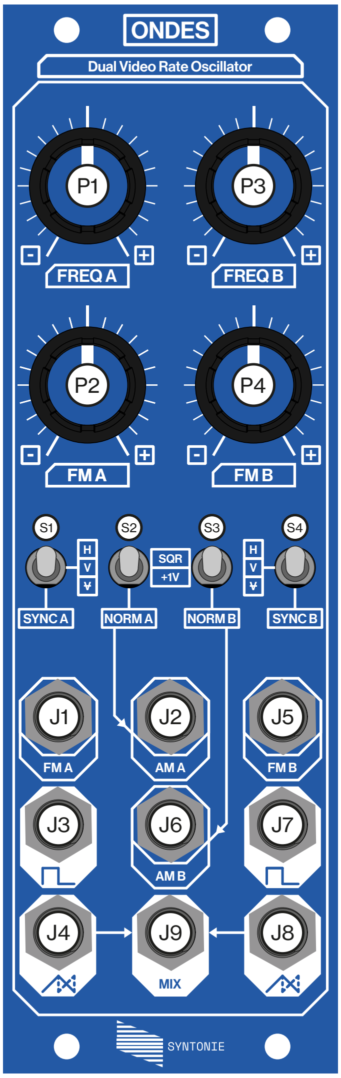
Ondes is a dual video rate oscillator, featuring two independent oscillators with selectable sync, frequency and amplitude modulation, square and triangle/sawtooth wave outputs, as well as a mix output, offering a lot of versatility in pattern creation.

Specifications

– 8HP
– 200 mA +12V (16pin or DC)
– 0 mA -12V
– 0 mA +5V
– 42mm depth

Special thanks to: **LZX team** for the Cadet Oscillator design, which has been the starting point of this module.
Lorenzo Ferronato for the documentation design // And of course, **everyone who has supported Syntonie until now & those who will support it in the future.**

Interface



(P1) Oscillator A frequency manual control

(P2) Oscillator A frequency CV attenuverter

(S1) Oscillator A sync selection switch

(S2) Oscillator A AM normalization switch

(J1) Oscillator A frequency CV input: 0-1V, 100kΩ, minijack

(J2) Oscillator A amplitude CV input: 0-1V, 100kΩ, minijack

(J3) Oscillator A square wave output: 0-1V, 75Ω, minijack

(J4) Oscillator A tri/saw wave output: 0-1V, 75Ω, minijack

(P3) Oscillator B frequency manual control

(P4) Oscillator B frequency CV attenuverter

(S2) Oscillator B sync selection switch

(S3) Oscillator B AM normalization switch

(J5) Oscillator B frequency CV input: 0-1V, 100kΩ, minijack

(J6) Oscillator B amplitude CV input: 0-1V, 100kΩ, minijack

(J7) Oscillator B square wave output: 0-1V, 75Ω, minijack

(J8) Oscillator B tri/saw wave output: 0-1V, 75Ω, minijack

(J9) Oscillator A and B tri/saw wave output: 0-1V, 75Ω, minijack

Ondes features two identical oscillators, here is a description of each of its parameters:

Synchronisation and frequency range:

The synchronisation switch (**SYNC A, SYNC B**) acts both on the sync signal selection and frequency range:

- **H**: horizontal sync is selected, producing vertical bars, with a frequency ranging from 15kHz to 1.5MHz
- **V**: vertical sync is selected, producing horizontal bars, with a frequency ranging from 23.98Hz to 3.5kHz
- **~~V~~**: no sync is selected, the oscillator is free running, producing up/down scrolling horizontal bars, with a frequency ranging from 10Hz to 3.5kHz

Note: a valid sync signal needs to be present at the rear RCA sync input for the H and V switch positions to work as expected.

Frequency modulation:

The initial frequency of the oscillator is set using the manual frequency control (**FREQ A**, **FREQ B**).

The frequency can be modulated with another signal, by plugging it into the FM CV input (**FM A**, **FM B**) and turning the attenuverter clockwise/counter-clockwise.:

- When the attenuverter is turned fully clockwise, the CV signal is non-inverted at maximum modulation depth.
- When the attenuverter is turned fully counter-clockwise, the CV signal is inverted at maximum modulation depth.
- When the attenuverter is in center position, the CV is nulled and has no longer an effect on the frequency of the oscillator.

Note: when no signal is present at the FM CV input, the attenuverter can be used to reach the minimum/maximum frequencies of the oscillator. It is also handy in unsynced mode (**~~V~~**) as it allows to fine tune the frequency for a slower up/down scroll.

Amplitude modulation:

Ondes features amplitude modulation on the triangle wave output only, the square wave isn't affected by amplitude modulation. The amplitude modulation is achieved by the mean of a 4-quadrant multiplier, here is how it is configured:

- When the signal present at the amplitude modulation CV input (**AM A**, **AM B**) is at +1V, the triangle output is non-inverted at maximum amplitude.
- When the signal present at the amplitude modulation CV input is at 0V, the triangle output is inverted at maximum amplitude.
- When the signal present at the amplitude modulation CV input is at 0.5V, the triangle output is fully attenuated.

All values in-between 0.5V and 1V will result in a non-inverted triangle with varying amplitude.
All values in-between 0V and 0.5V will result in an inverted triangle with varying amplitude.

AM normalization switch:

The amplitude modulation CV input (**AM A, AM B**) are normalized to the amplitude modulation normalization switch (**NORM A**, **NORM B**) meaning that when no external signal is present at the AM input, the switch will provide the following signals:

- **+1V**: when selected, the triangle output is at max amplitude, non-inverted. This is the normal operation mode.
- **SQR**: when selected, the square wave output of the oscillator is routed into the AM input, producing a sawtooth wave at twice the frequency of the oscillator core frequency. When the square wave is low, the triangle wave is inverted, and when the square wave is high, the triangle wave is non-inverted, effectively inverting half of the triangle wave which result in a sawtooth wave.

Plugging a signal into the AM input will override this normalization, so the external signal now controls the amplitude of the triangle.

Outputs:

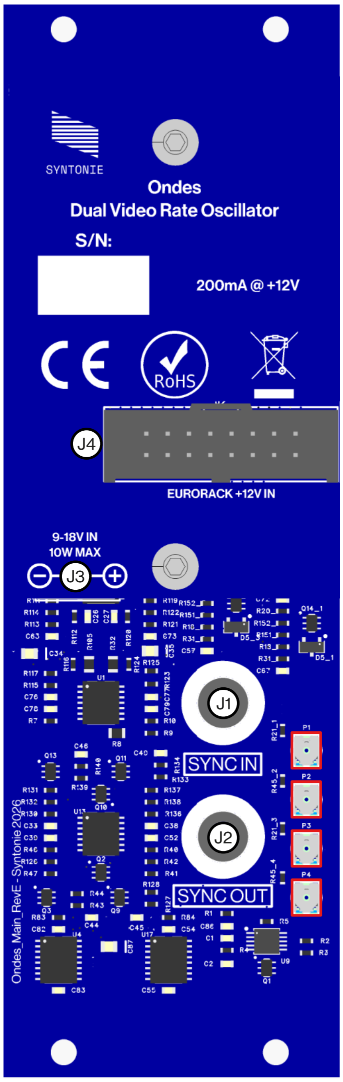
Each oscillator features a square wave output, going from 0V to 1V and 1V to 0V almost instantly (in fact with roughly 30ns transition times), which translate to a signal alternating from minimum to maximum brightness with hard transitions. The square wave isn't affected by the amplitude modulation.

Ondes also features a linear wave output, which gives a triangle wave when the AM normalization switch (**NORM A**, **NORM B**) is set to **+1V**, and a sawtooth wave when the switch is in **SQR** position, which translates to a signal alternating from minimum to maximum brightness with a smooth transition, looking like a gradient. Since the sawtooth is produced by the mean of amplitude modulation, its amplitude cannot be modulated.

The **MIX** output is the sum of oscillator A and oscillator B linear waveform output, its typical use case is to produce a 2D pattern when one oscillator is set to H and the other oscillator is set to V.

- When both oscillators are set to output triangle waves, the resulting sum is a diamond shape
- When one oscillator is set to triangle and the other to sawtooth, the resulting sum is a triangle shape
- When both oscillators are set to output sawtooth waves, the resulting sum is a diagonal gradiant

Rear sync connections and trimmers



(J1) Synchronisation input: 0V-300mV, 75Ω, RCA

(J2) Synchronisation input: 0V-300mV, 75Ω, RCA

(J3) DC power input: +12VDC, 5.5mm x 2.1mm DC barrel

(J4) Eurorack power input: +12VDC, 16pin IDC connector

(P1) Positive triangle A offset trimmer

(P2) Negative triangle A offset trimmer

(P3) Positive triangle B offset trimmer

(P4) Negative triangle B offset trimmer

Notes: In a setup based around an encoder/sync generator (**Sortie**), the sync output of the sync generator is connected to the sync input of Ondes directly or Ondes can be placed anywhere in the sync chain.

In Composite video processing setup (**Stable**, **Détail**), the sync output of the video processor is typically connected to the sync input of Ondes directly or Ondes can be placed anywhere in the sync chain.

Notes: the trimmers present on Ondes main board controls the offsets at the output of the four quadrant multipliers used for amplitude modulation. It allows to correct for offset errors, which could be particularly visible when the linear output is set to output a sawtooth.

The module comes already calibrated, however if the sawtooth brightness varies from too much from one cycle to the other, you can adjust the negative triangle offset trimmer (**P2** for oscillator A, **P4** for oscillator B) so the brightness matches better.

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